A compressed SDP dual bound inside MISDP branch and bound

YALMIP autoresearch note

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Abstract

Some MISDP instances contain ordered binary thresholds whose SDP columns are almost identical. Solving an SDP with one level variable per group is cheap, but its objective is not necessarily a lower bound for the original problem. The implemented BNB method uses that small SDP only to obtain a positive semidefinite dual direction and a branching point. It computes any pruning bound from the original node coefficients.

The bound

At a BNB node, write the original relaxation as

$$\min_{x\in P_{\mathrm{lin}}} f(x) = f_0 + c^{\mathsf{T}}x \quad \text{subject to} \quad S(x) = C - \sum_i A_i x_i \succeq 0, \tag{1}$$

where P_{lin} contains the original linear rows and variable bounds. When threshold variables $y_1 \geq \cdots \geq y_k$ have nearly equal SDP columns, an auxiliary SDP replaces them by a bounded level $\ell = \sum_i y_i$. Its coefficients may only approximate those in (1). We therefore use its solution as follows:

1. Extract an SDP dual matrix, project it to a Gram factor B, and set $Z = B^{\mathsf{T}}B \succeq 0$.

2. Compute
$$L(Z) = \min_{x\in P_{\mathrm{lin}}} \big[f(x) - \langle Z, S(x)\rangle\big]. \tag{2}$$
Every feasible x has $\langle Z, S(x)\rangle \geq 0$, so $L(Z)$ is a lower bound for the *original* node regardless of errors in the auxiliary SDP. This minimization is a linear program.

3. Check the LP dual on the original rows. The implementation clips its multipliers to nonnegative values and minimizes any remaining stationarity residual over outward-rounded variable boxes. Error allowances cover the Gram product and scalar products. If a finite bound cannot be established, solve the original SDP node.

The auxiliary primal point may choose a fractional original threshold for branching. It cannot become an incumbent: an integral candidate receives an exact original-SDP node solve. The root is also solved exactly.

A two-threshold example

Let $\delta = 0.02$ and consider

$$\begin{aligned} \min \quad & y_1 + y_2 \\ \text{subject to} \quad & y_2 \le y_1, \qquad y_1, y_2 \in \{0,1\}, \\ & S_\delta(y) = \begin{pmatrix} 1 & 1/2 \\ 1/2 & y_1 + (1+\delta)y_2 \end{pmatrix} \succeq 0. \end{aligned} \tag{3}$$

The binary optimum is 1, attained at $(y_1, y_2) = (1, 0)$. In the continuous relaxation, positive semidefiniteness requires $y_1 + (1+\delta)y_2 \ge 1/4$. Because $y_2 \le y_1$, its optimum is

$$v_{\text{original}}^{\text{LP/SDP}} = \frac{1/4}{1+\delta/2} = 0.24752475\ldots. \tag{4}$$

Now let $\ell = y_1 + y_2$ and replace the bottom-right entry by ℓ in an auxiliary compressed SDP. Its optimum is $\ell = 1/4$. Using this 0.25 as an original lower bound would be *invalid*: it exceeds (4). The compressed SDP supplies the Gram factor

$$B = \begin{pmatrix} 1/2 & -1 \end{pmatrix}, \qquad Z = B^{\mathsf{T}} B = \begin{pmatrix} 1/4 & -1/2 \\ -1/2 & 1 \end{pmatrix}. \tag{5}$$

For the *original* matrix, $\langle Z, S_\delta(y) \rangle = y_1 + (1+\delta)y_2 - 1/4$. Hence (2) becomes

$$L(Z) = \min_{0 \le y_2 \le y_1 \le 1} \left(\frac{1}{4} - \delta y_2 \right) = \frac{1}{4} - \delta = 0.23 \ \le \ 0.24752475\ldots. \tag{6}$$

The correction loses a little strength, but it restores a valid lower bound. In a BNB node it can justify pruning only when it exceeds the incumbent by the prescribed numerical margin; otherwise the compressed point is used for branching or the original SDP is solved.

Observed performance and scope

In one paired ten-model truss panel, exact BNB used 3,090 nodes and 37.56 seconds, versus 2,842 nodes and 28.47 seconds with the checked compressed path. All paired objectives agreed. The full BNB test suite had the same passes and failures with the path enabled and disabled; suite times varied between alternating runs. The method is enabled only for detected level groups with a supported single SDP block, linear objective, and Mosek lower solver. Extreme numeric scales and failed certificate checks fall back to the original SDP.